

Centre No.						Paper Reference							Surname	Initial(s)	
Candidate No.						6	6	7	7	/	0	1	R	Signature	

Paper Reference(s)

6677/01R

Edexcel GCE

Mechanics M1

Advanced/Advanced Subsidiary

Friday 6 June 2014 – Afternoon

Time: 1 hour 30 minutes

Examiner's use only

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Team Leader's use only

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Materials required for examination

Mathematical Formulae (Pink)

Items included with question papers

Nil

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation or symbolic differentiation/integration, or have retrievable mathematical formulae stored in them.

Instructions to Candidates

In the boxes above, write your centre number, candidate number, your surname, initials and signature. Check that you have the correct question paper.

Answer ALL the questions.

You must write your answer to each question in the space following the question.

Whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either two significant figures or three significant figures.

When a calculator is used, the answer should be given to an appropriate degree of accuracy.

Information for Candidates

A booklet 'Mathematical Formulae and Statistical Tables' is provided.

Full marks may be obtained for answers to ALL questions.

The marks for individual questions and the parts of questions are shown in round brackets: e.g. (2).

There are 7 questions in this question paper. The total mark for this paper is 75.

There are 28 pages in this question paper. Any blank pages are indicated.

Advice to Candidates

You must ensure that your answers to parts of questions are clearly labelled.

You should show sufficient working to make your methods clear to the Examiner.

Answers without working may not gain full credit.

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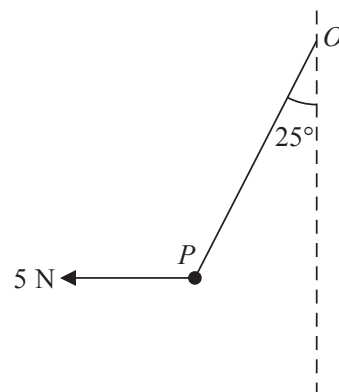


Figure 1

A particle P of weight W newtons is attached to one end of a light inextensible string. The other end of the string is attached to a fixed point O . A horizontal force of magnitude 5 N is applied to P . The particle P is in equilibrium with the string taut and with OP making an angle of 25° to the downward vertical, as shown in Figure 1.

Find

- (a) the tension in the string,

(3)

- (b) the value of W .

(3)



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Question 1 continued



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- (a) Find the value of q .

(4)

At time $t = 0$, P is moving with velocity $(-2\mathbf{i} + 4\mathbf{j}) \text{ m s}^{-1}$.

- (b) Find the speed of P at time $t = 2$ seconds.

(6)

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3. A car starts from rest and moves with constant acceleration along a straight horizontal road. The car reaches a speed of $V \text{ m s}^{-1}$ in 20 seconds. It moves at constant speed $V \text{ m s}^{-1}$ for the next 30 seconds, then moves with constant deceleration $\frac{1}{2} \text{ m s}^{-2}$ until it has speed 8 m s^{-1} . It moves at speed 8 m s^{-1} for the next 15 seconds and then moves with constant deceleration $\frac{1}{3} \text{ m s}^{-2}$ until it comes to rest.

(a) Sketch, in the space below, a speed-time graph for this journey.

(3)

In the first 20 seconds of this journey the car travels 140 m.

Find

(b) the value of V ,

(2)

(c) the total time for this journey,

(4)

(d) the total distance travelled by the car.

(4)



Question 3 continued



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4. At time $t = 0$, a particle is projected vertically upwards with speed u from a point A . The particle moves freely under gravity. At time T the particle is at its maximum height H above A .

- (a) Find T in terms of u and g . (2)

- (b) Show that $H = \frac{u^2}{2g}$ (2)

The point A is at a height $3H$ above the ground.

- (c) Find, in terms of T , the total time from the instant of projection to the instant when the particle hits the ground.
- (4)**

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Question 4 continued



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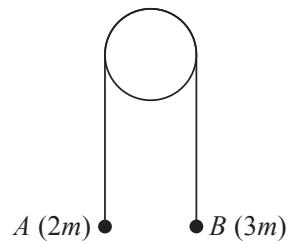


Figure 2

Two particles A and B have masses $2m$ and $3m$ respectively. The particles are connected by a light inextensible string which passes over a smooth light fixed pulley. The system is held at rest with the string taut. The hanging parts of the string are vertical and A and B are above a horizontal plane, as shown in Figure 2. The system is released from rest.

- (a) Show that the tension in the string immediately after the particles are released is $\frac{12}{5}mg$.

After descending 1.5 m, B strikes the plane and is immediately brought to rest. In the subsequent motion, A does not reach the pulley.

- (b) Find the distance travelled by A between the instant when B strikes the plane and the instant when the string next becomes taut.
- (6)

Given that $m = 0.5$ kg,

- (c) find the magnitude of the impulse on B due to the impact with the plane. (2)



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Question 5 continued



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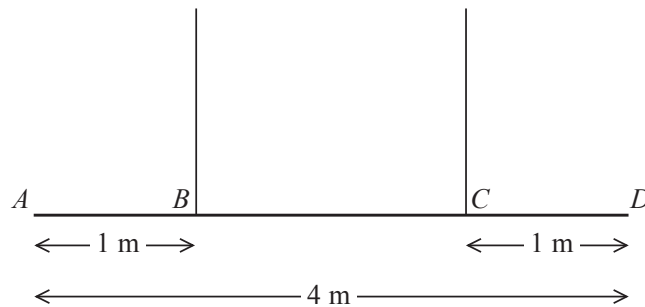


Figure 3

A non-uniform beam AD has weight W newtons and length 4 m. It is held in equilibrium in a horizontal position by two vertical ropes attached to the beam. The ropes are attached to two points B and C on the beam, where $AB = 1$ m and $CD = 1$ m, as shown in Figure 3. The tension in the rope attached to C is double the tension in the rope attached to B . The beam is modelled as a rod and the ropes are modelled as light inextensible strings.

- (a) Find the distance of the centre of mass of the beam from A . (6)

A small load of weight kW newtons is attached to the beam at D . The beam remains in equilibrium in a horizontal position. The load is modelled as a particle.

Find

- (b) an expression for the tension in the rope attached to B , giving your answer in terms of k and W ,
- (3)**

- (c) the set of possible values of k for which both ropes remain taut. (2)

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Question 6 continued



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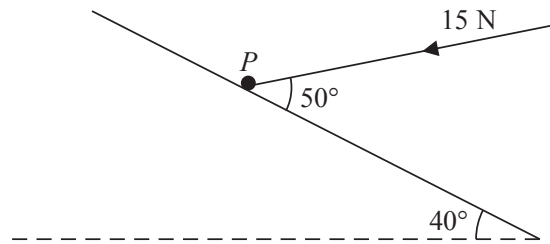


Figure 4

A particle P of mass 2.7 kg lies on a rough plane inclined at 40° to the horizontal. The particle is held in equilibrium by a force of magnitude 15 N acting at an angle of 50° to the plane, as shown in Figure 4. The force acts in a vertical plane containing a line of greatest slope of the plane. The particle is in equilibrium and is on the point of sliding down the plane.

Find

- (a) the magnitude of the normal reaction of the plane on P ,
(b) the coefficient of friction between P and the plane.

The force of magnitude 15 N is removed.

- (c) Determine whether P moves, justifying your answer. (4)

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Q7

TOTAL FOR PAPER: 75 MARKS

28

